

Pelycosaur

Mammals evolved from a group of reptilelike animals called pelycosaur. The pelycosaur lived long before even the dinosaurs and for a while were the largest animals on land. They looked more like lizards than mammals, but their link with mammals is clear from a special hole in the skull behind each eye. As in mammals, the jaw muscles passed through this hole, giving these animals a killer bite.



FAMILY FACT FILE

Key features

- Cold-blooded
- Reproduced by laying eggs
- Small brains
- Lizardlike, sprawling legs
- Short claws on toes
- Holes in the skull behind the eyes
- Varied teeth

When

Pelycosaur first appeared in the Late Carboniferous, 320 million years ago. They died out in the Late Permian, 251 million years ago.

Dimetrodon

die-MET-roe-don



- **When** 280 million years ago (Early Permian)
- **Fossil location** Germany, USA
- **Habitat** Swamps
- **Length** 10 ft (3 m)
- **Diet** Meat

Dimetrodon was the most fearsome predator of its time. It was built like a Komodo dragon, but with a huge “sail” on its back formed from skin wrapped over rods of bone. *Dimetrodon* means “two-sized tooth”—unlike most reptiles, which have teeth that are similar to each other, *Dimetrodon* had teeth of several types, as mammals have. At the front of the mouth were long, daggerlike canines for piercing and grabbing flesh; at the back were smaller, sharp-edged teeth for slicing flesh.



▲ **FOSSIL TRACKS** *These five-toed footprints may have been left by Dimetrodon, one of the most common animals of its time.*

Caninelike fangs

Dimetrodon had a hole in the skull behind each eye socket. Strong jaw muscles went through these holes, giving it a powerful bite. Humans share the same feature.

